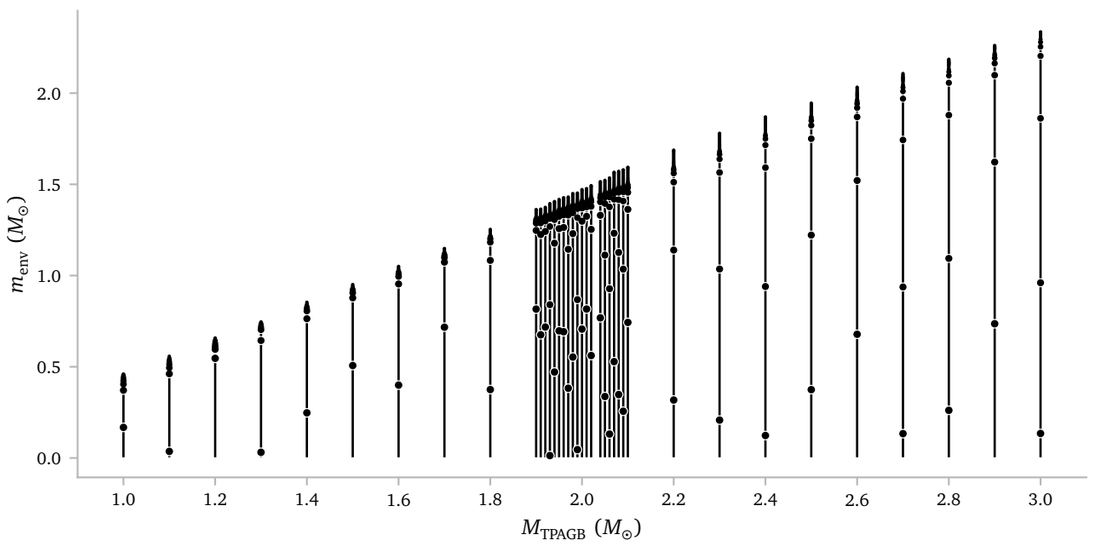
Koen Essers • Master Internship • November 11, 2026

Notes Week 36

a. Dense Mass Spacing Results

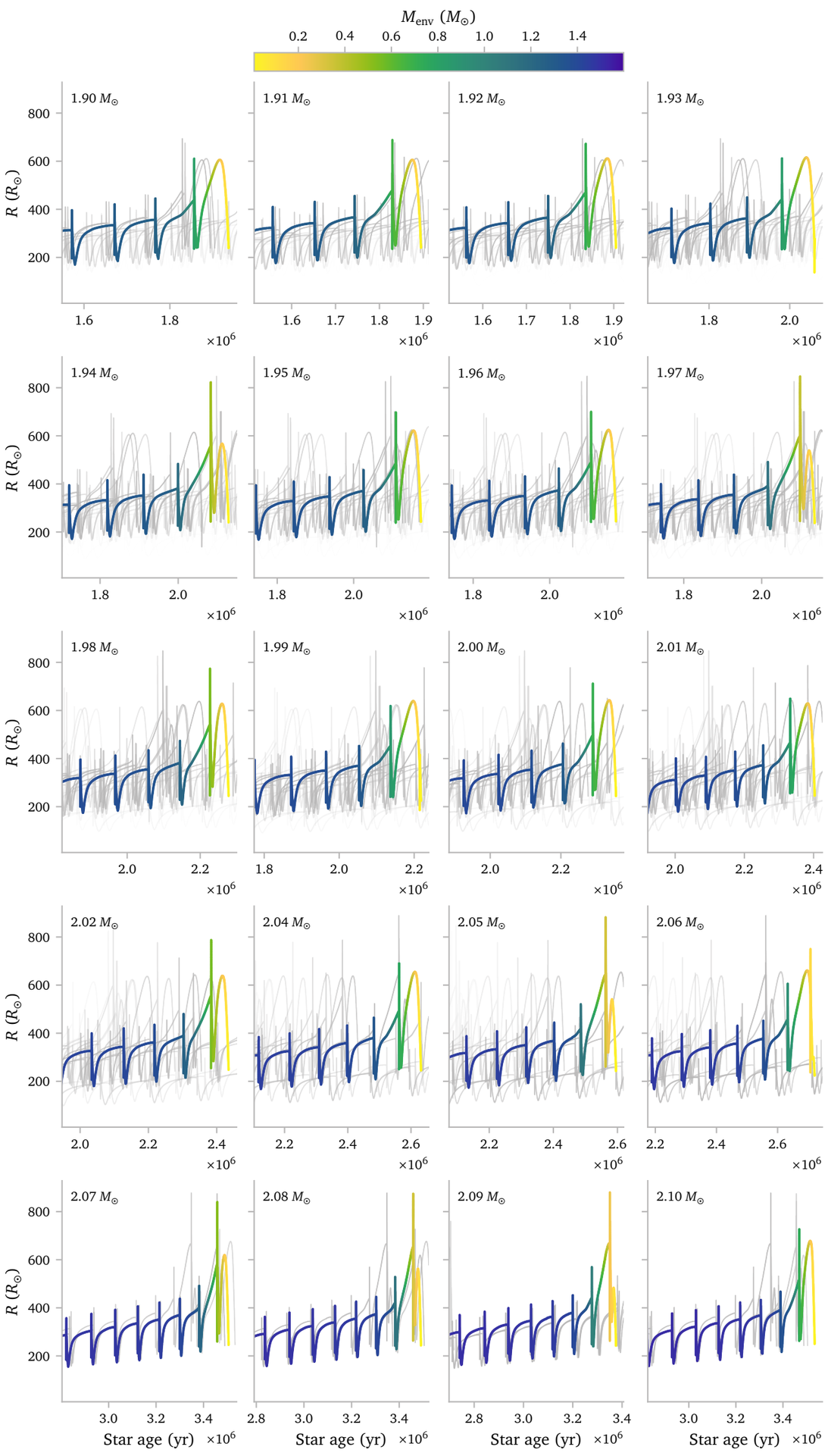
Below I plot the envelope mass for the new models and the old ones.



Quite clearly, even for 0.01 $M_\odot$ differences in mass, there is no predictable relation between changes in initial mass and envelope mass during the last thermal pulse.

It seems to be very "chaotic". Small perturbations in initial mass lead to quite drastically different final TPAGB evolutions.

To highlight this, let me show the last few thermal pulses for the different masses between 1.9 $M_\odot$ and 2.1 $M_\odot$.

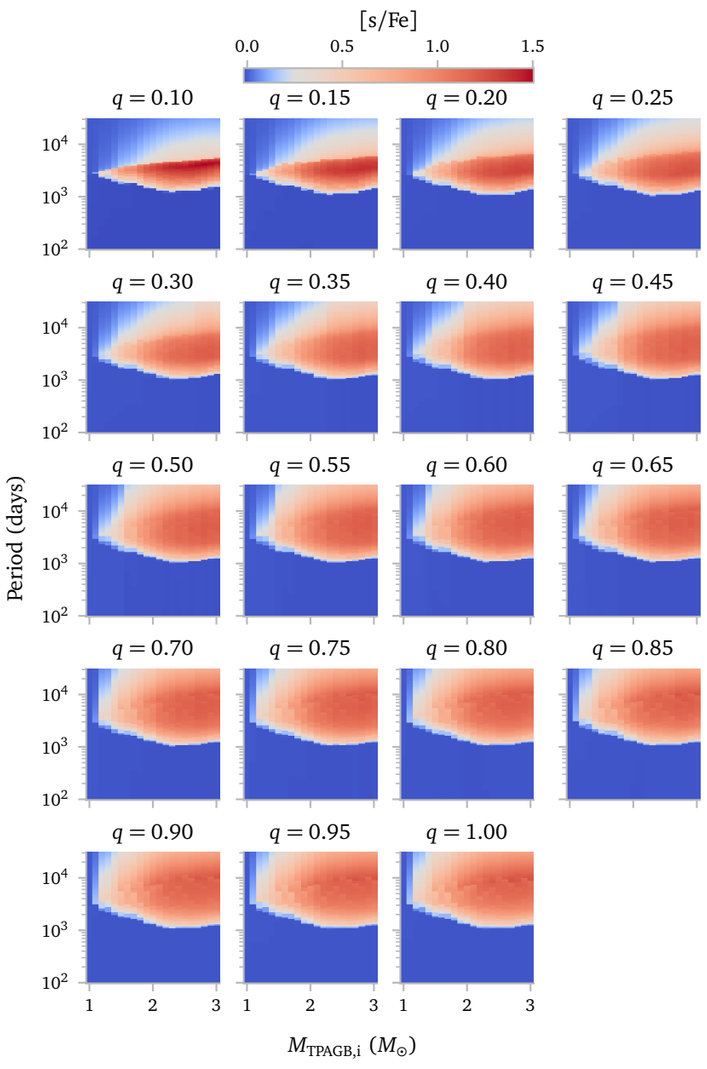


Clearly, we see that these very small mass differences can have quite different evolutions. For example, look at the systems with masses 1.93 $M_\odot$, 1.94 $M_\odot$ and 1.95 $M_\odot$.

b. Simple Binary grids

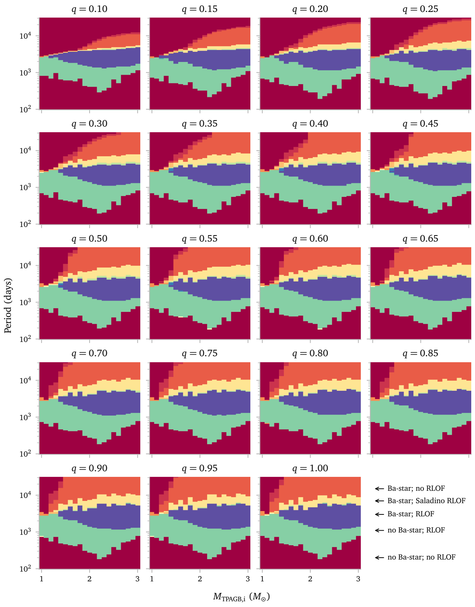
I simulated a bunch of systems using the Simple binary code and my single stars. I used the three different wind prescriptions to compare what parts of the parameter space barium dwarfs produced through RLOF take up.

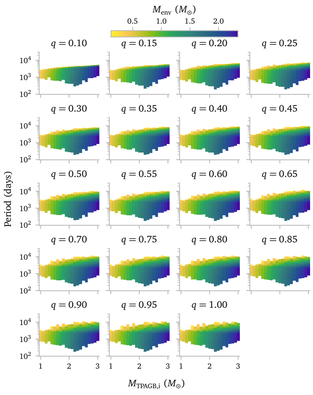
I decided to simulate the models based on initial period instead of initial Roche lobe radius. This is only because this will make it easier to interpolate the *many* Moe & Di Stefano (2017) systems I will generate. Obviously it is trivial to convert them back any time.



The figure shows the [s/Fe] values of the accretor as predicted by the simple binary evolution code using $\varepsilon_{\rm RLOF} = 0.1$ and the Saladino et al. (2018) winds.

The figure below uses this same $\varepsilon_{\rm RLOF} = 0.1$. I use the criterion of [s/Fe] > 0.25 for a barium star.

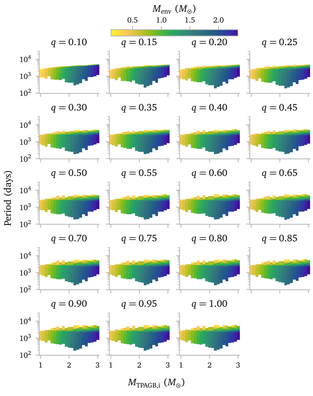




Here, I color *all* systems that undergo RLOF during the TPAGB (independent of *if* they produce a barium star or not) based on the final envelope mass at the point of first RLOF.

Unsurprisingly, for short initial periods, there is no difference if you change the period a bit. This is to be expected because the stars do not experience much wind in this regime. You clearly see the *full* envelope mass still intact at the moment of RLOF.

If you move to higher initial periods, you get into the regime where winds start to matter and it is possible to undergo RLOF with very little envelope mass left, even though the star was initially quite massive.



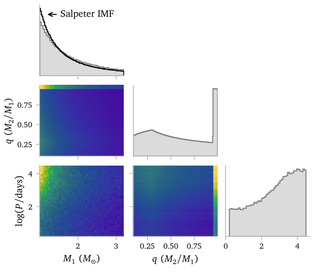
This figure is the same as the figure above, except now I use the Bondi (1952) Hoyle & Lyttleton (1939) prescription.

If we compare this figure with the one using Saladino et al. (2018) winds, we see that the BHL prescription mainly prevents systems from interacting with decreased envelope masses. This makes sense as clearly this is where the winds become important.

c. Moe and Di Stefano

I use a standard Salpeter (1955) IMF. Differences between this IMF and for example, Chabrier (2003) should be very minor, since we are only interested in primaries that produce enough s-process elements in third dredge-up episodes. This happens at masses of about 1.2 $M_\odot$ and higher. We also only go up to masses of $M = 3.0\,M_\odot$. Above this mass, it is difficult to reproduce barium dwarfs with masses within the observed range. Therefore, I have not simulated these TPAGB stars with MESA for masses above 3.0 $M_\odot$.

For the plots below, I only show periods below 30 000 days, because periods above this limit will not produce barium dwarfs within the observed period range, and *definitely* not RLOF TPAGB binaries.

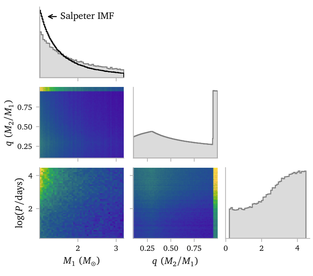


Here, I show a population of 153714(000) *binaries* using the parameterization of Moe & Di Stefano (2017). Their paper explicitly describes multiplicities, and not only binaries, so really I generated a bunch of singles, binaries, triples and quadruples, but I only show the binaries here.

Clearly, the mass distribution is similar to the Salpeter (1955) IMF. However, it is slightly flatter. There are slightly more heavier stars and slightly fewer lighter stars.

The binaries also contain many "twin" binaries. This is about 10 to 15% of the total systems.

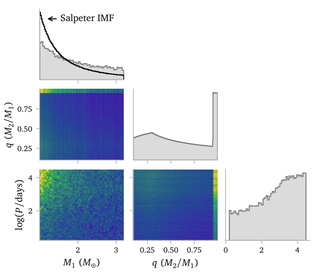
Lastly, we see that, in log space, there are many more wide orbits. However, if you would look at the systems in linear space, the peak is towards lower periods.



Here, I show the same distributions, but now for the population containing 91111(000) triples.

There are fewer triples, which is to be expected within this mass range. Let me be fully clear on what I do with the triples as well as the quadruples. Triples contain two binaries. I simply include none, one, or both of these orbits, only dependent on if they are within the period range. This means the fraction of triple *orbits* over binary *orbits* will be *larger* than the fraction of triple *systems* over binary *systems*.

The mass-ratio distribution and period distributions are not very different. However, we see a much shallower IMF.

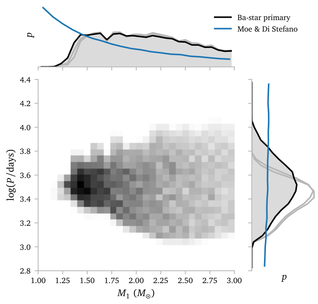


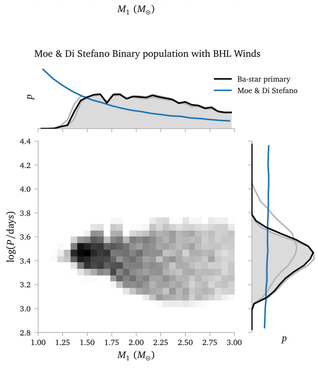
Here, I show the same distributions, but now for the population containing 27730(000) triples.

The mass-ratio distribution and period distributions are not very different. However, we see a much shallower IMF, even shallower than for the triples.

For the figures below, I use $\varepsilon_{\rm RLOF} = 0.1$. I also classify accretors as barium stars if they have [s/Fe] > 0.25.

Moe & Di Stefano Binary population with Saladino Winds

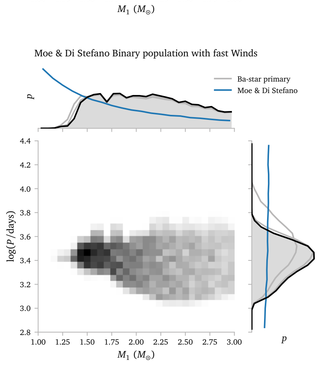




This figure shows the density of binaries generated from the Moe & Di Stefano (2017) distributions. Only binaries that are barium stars *and* undergo RLOF during the TPAGB are counted. The systems are assumed to experience Saladino et al. (2018) style winds.

I plot the primary mass and period distributions separately as well. They are clearly different from the total distribution.

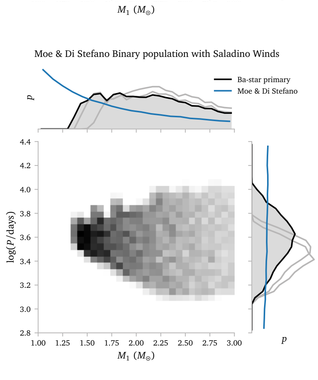
Moe & Di Stefano Binary population with BHL Winds



This figure shows the density of binaries generated from the Moe & Di Stefano (2017) distributions. Only binaries that are barium stars *and* undergo RLOF during the TPAGB are counted. The systems are assumed to experience Bondi (1952) Hoyle & Lyttleton (1939) style winds.

Compared to the Saladino winds, this population has shorter initial periods. The mass distribution is similar.

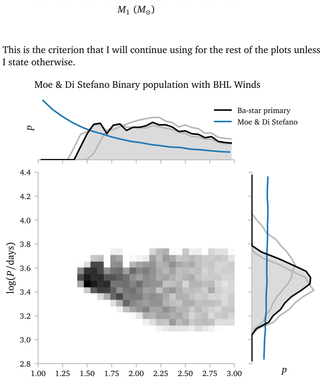
Moe & Di Stefano Binary population with fast Winds



This figure shows the density of binaries generated from the Moe & Di Stefano (2017) distributions. Only binaries that are barium stars *and* undergo RLOF during the TPAGB are counted. The systems are assumed to experience fast winds.

This population is very similar to the Bondi (1952) Hoyle & Lyttleton (1939) style winds population.

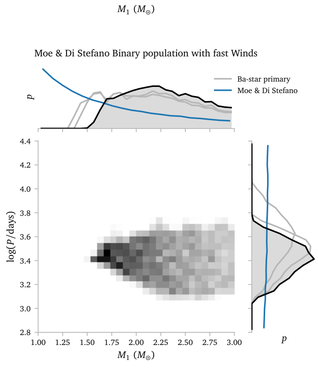
Moe & Di Stefano Binary population with Saladino Winds



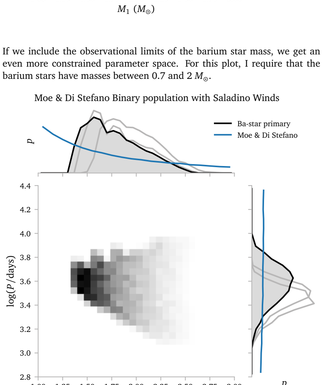
If we require the binaries to have $q > 0.5$ for stable mass transfer, we get slightly different distributions. Here I also increased the barium threshold to [s/Fe] > 0.5.

This is the criterion that I will continue using for the rest of the plots unless I state otherwise.

Moe & Di Stefano Binary population with BHL Winds



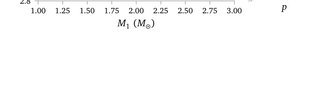
Moe & Di Stefano Binary population with fast Winds



Something interesting happens here. We see that we lose the low mass part of the systems. This makes sense. Fast winds do not accrete anything and the RLOF alone is no longer able to produce the barium stars for these masses.

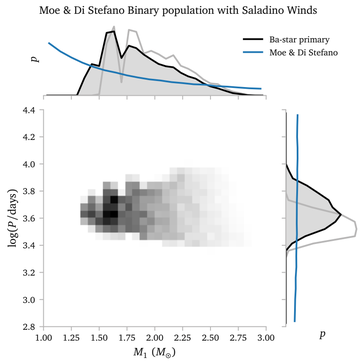
If we include the observational limits of the barium star mass, we get an even more constrained parameter space. For this plot, I require that the barium stars have masses between 0.7 and 2 $M_\odot$.

Moe & Di Stefano Binary population with Saladino Winds



$\varepsilon = 0.1$

Clearly, we lose most of the high mass systems, only systems with low q, but larger than 0.5 are still able to generate barium stars.

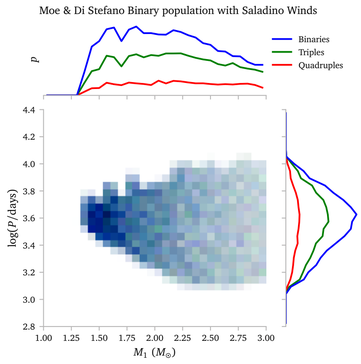


$\varepsilon = 0.0$

The fast winds models cannot produce barium stars. This is simply because we *only* have wind mass transfer, and the fast winds do not transfer mass.

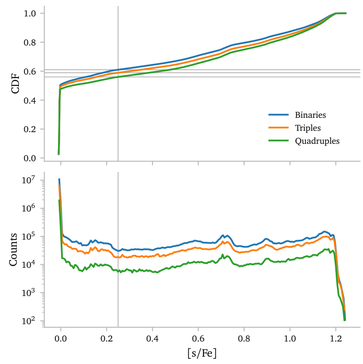
For the Saladino winds, we also see that the lower initial periods are now missing. This is because these models do not experience wind before RLOF occurs. In the simple case explored here, mass is stripped in its entirety during RLOF. However, we have seen that this is not very realistic. Therefore, the plot above is probably much more relevant.

Now I remove the mass criterion again. It is probably only an observational constraint anyway.

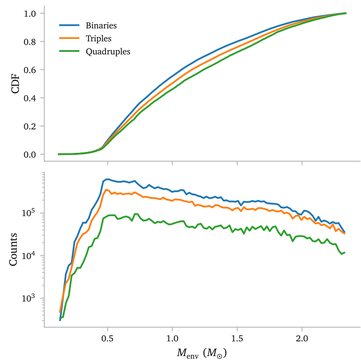


This plot shows the fraction of binaries, triples and quadruples. The colors in the 2d histogram give an idea of the fractions. If a square is blue, there are only binaries, if it is green there are only triples and if it is red, there are only quadruples. I white square contains all in equal proportions and a black square contains none. We see that the distribution is mostly blue, but towards higher masses, it gets a bit greener, meaning we have more triples there. The 1d histograms show this as well.

Below, I look at systems that undergo RLOF during the TPAGB.

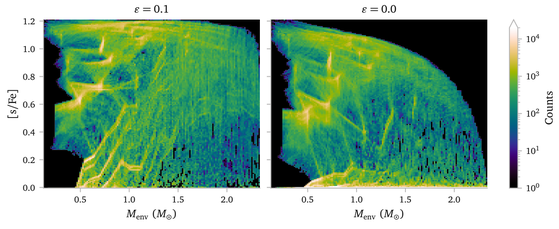


This figure shows how the [s/Fe] abundances are distributed. Clearly, many systems are not enriched at all. However, the rest of the abundances are quite evenly distributed.

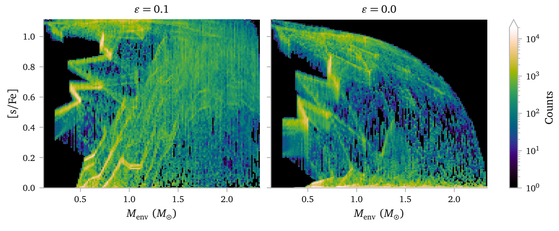


This figure shows how the envelope masses are distributed.

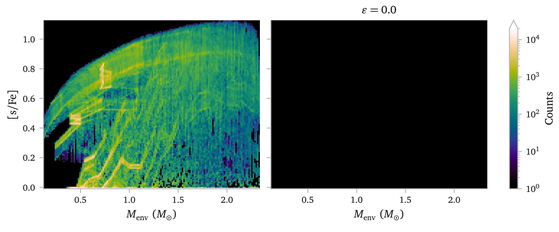
This figure shows a 2d histogram of the binaries from the plots above for Saladino style winds.



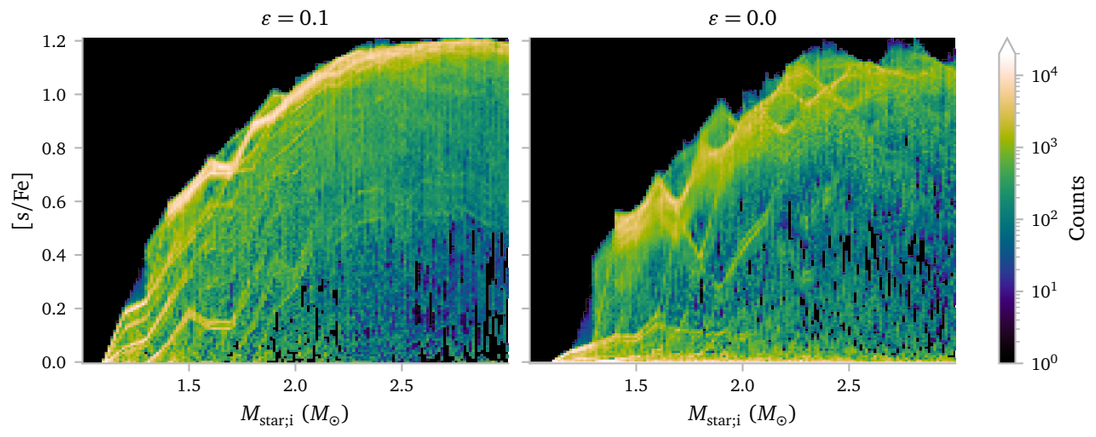
This figure shows a 2d histogram of the binaries from the plots above for BHL style winds.



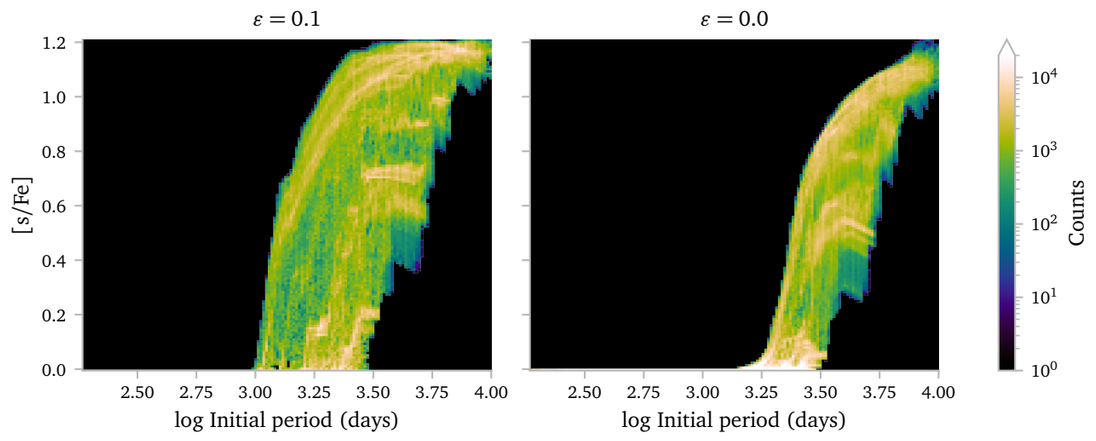
This figure shows a 2d histogram of the binaries from the plots above for fast style winds.



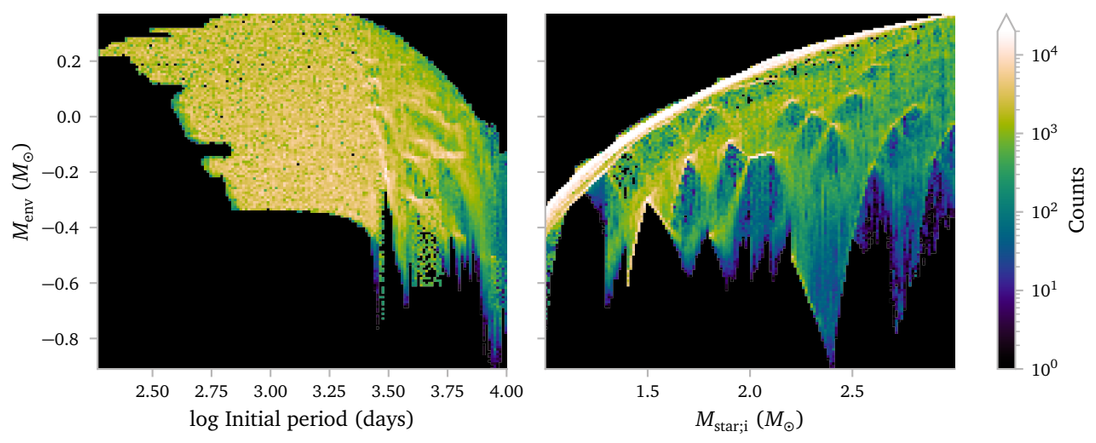
Now we look at [s/Fe] as a function of mass for Saladino style winds.



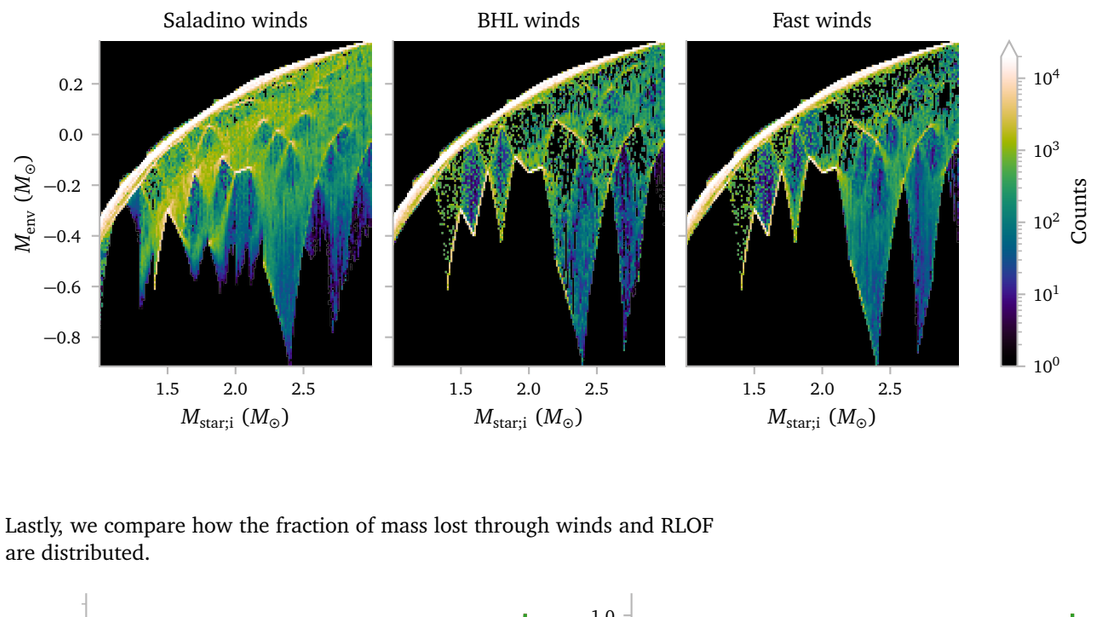
Now we look at [s/Fe] as a function of initial period for Saladino style winds.



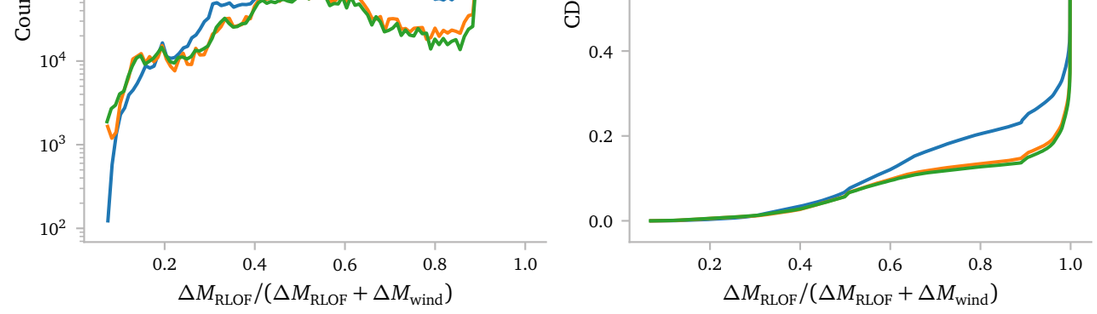
Now we look at M_{env} as a function of initial period and mass for Saladino style winds.

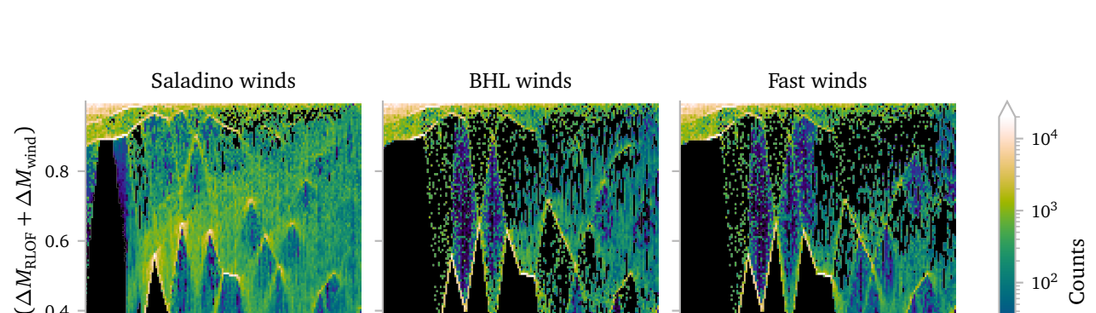


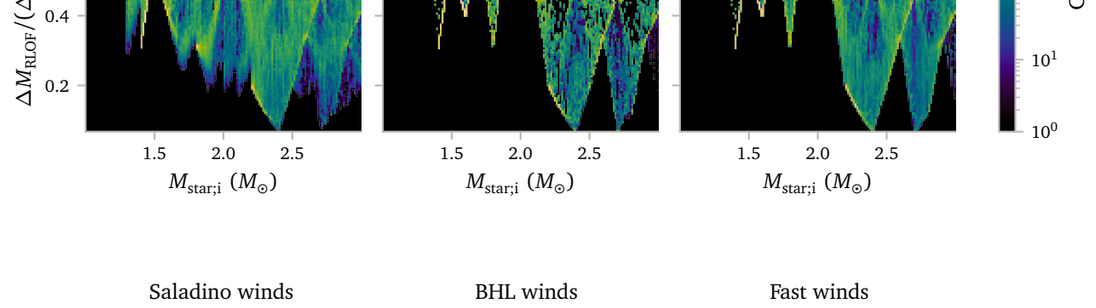
Now we look at M_{env} as a function of mass for the different style winds.



Lastly, we compare how the fraction of mass lost through winds and RLOF are distributed.







References

Bondi, H. (1952) *On spherically symmetrical accretion*. MNRAS112, 195.

Chabrier, G. (2003) *Galactic Stellar and Substellar Initial Mass Function*. PASP115, 763.

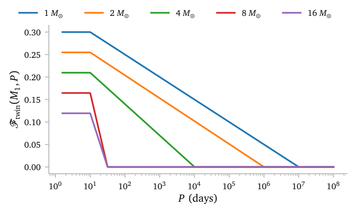
Hoyle, F. & Lyttleton, R. A. (1939) *The effect of interstellar matter on climatic variation*. Proceedings of the Cambridge Philosophical Society35, 405.

Moe, M. & Di Stefano, R. (2017) *Mind Your Ps and Qs: The Interrelation between Period (P) and Mass-ratio (Q) Distributions of Binary Stars*. ApJS230, 15.

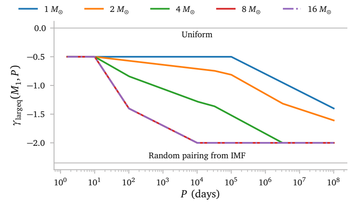
Saladino, M. I., Pols, O. R., van der Helm, E., Pelupessy, I., & Portegies Zwart, S. (2018) *Gone with the wind: the impact of wind mass transfer on the orbital evolution of AGB binary systems*. A&A618, A50.

Salpeter, E. E. (1955) *The Luminosity Function and Stellar Evolution.*. ApJ121, 161.

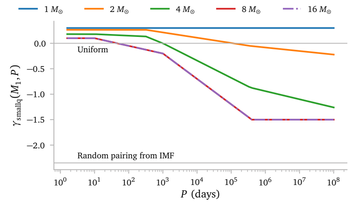
A Moe and Di Stefano Tests



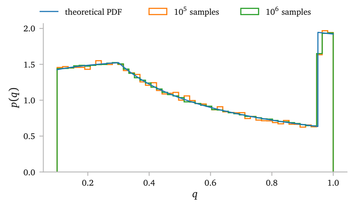
Compared with Figure 35 from Moe & Di Stefano (2017), these results look sensible.



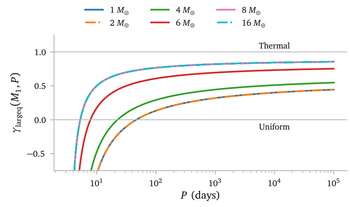
Compared with Figure 35 from Moe & Di Stefano (2017), these results look sensible.



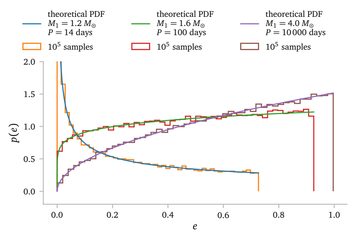
Compared with Figure 35 from Moe & Di Stefano (2017), these results look sensible.



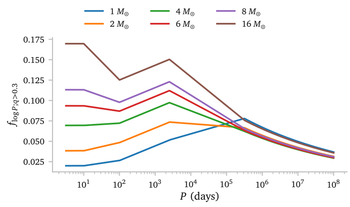
Compared with Figure 2 from Moe & Di Stefano (2017), these results look sensible.



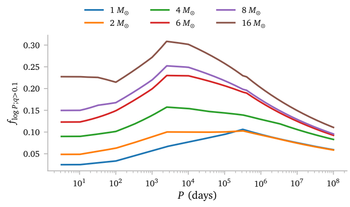
Compared with Figure 36 from Moe & Di Stefano (2017), these results look sensible.



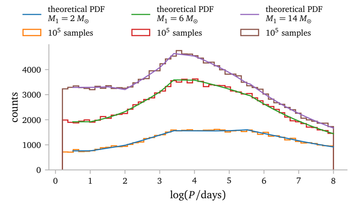
Based on how power laws should look and the figure above, I think this makes sense.



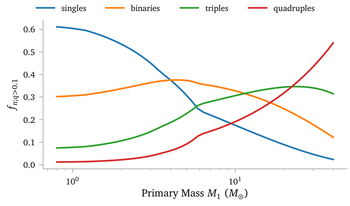
Compared with Figure 37 from Moe & Di Stefano (2017), these results look sensible.



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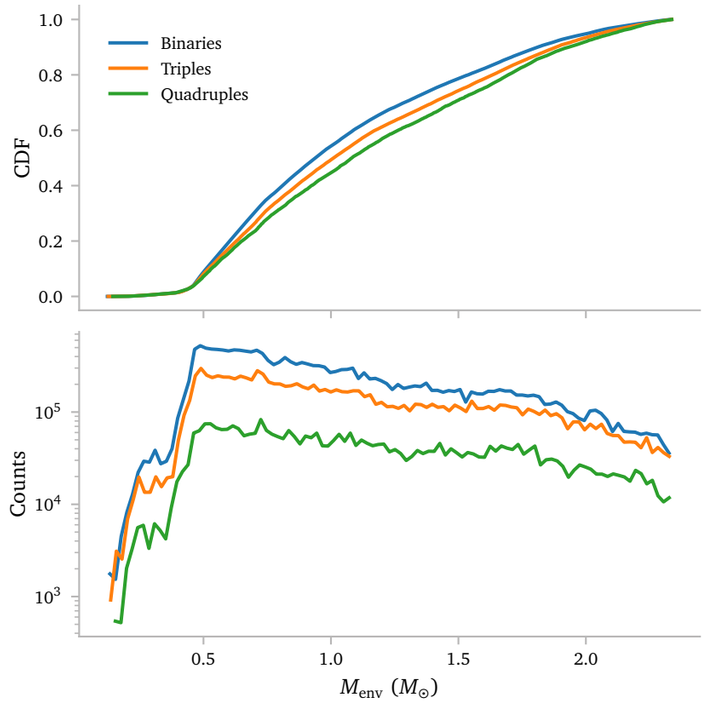


Sampling reproduces the period frequencies well.

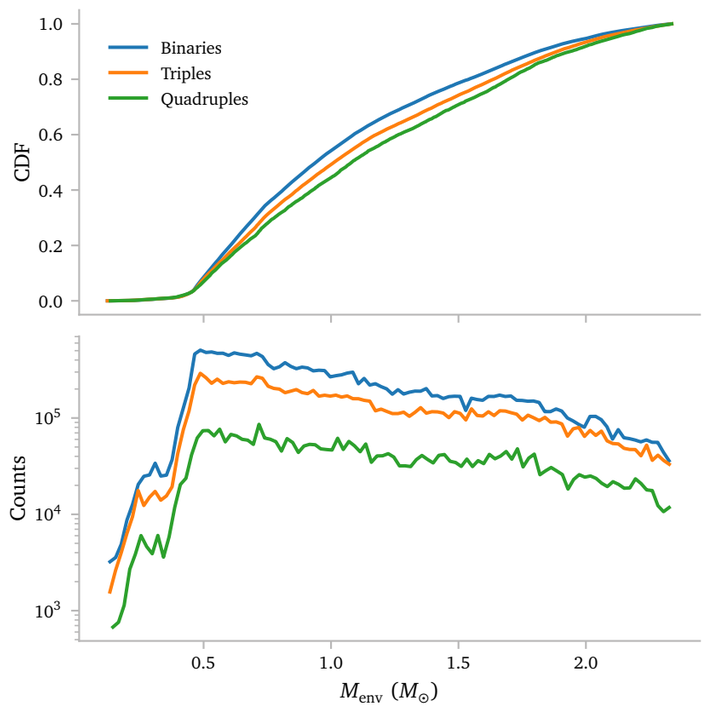


Compared with Figure 39 from Moe & Di Stefano (2017), these results look sensible.

B Additional Plots



Binaries, $\varepsilon = 0.1$, BHL



Binaries, $\varepsilon = 0.1$, fast